Lab 5 – Forward Kinematics

1 Introduction

In this lab the forward kinematics of a SCARA manipulator will be implemented in C++ and simulated.

2 Experiment

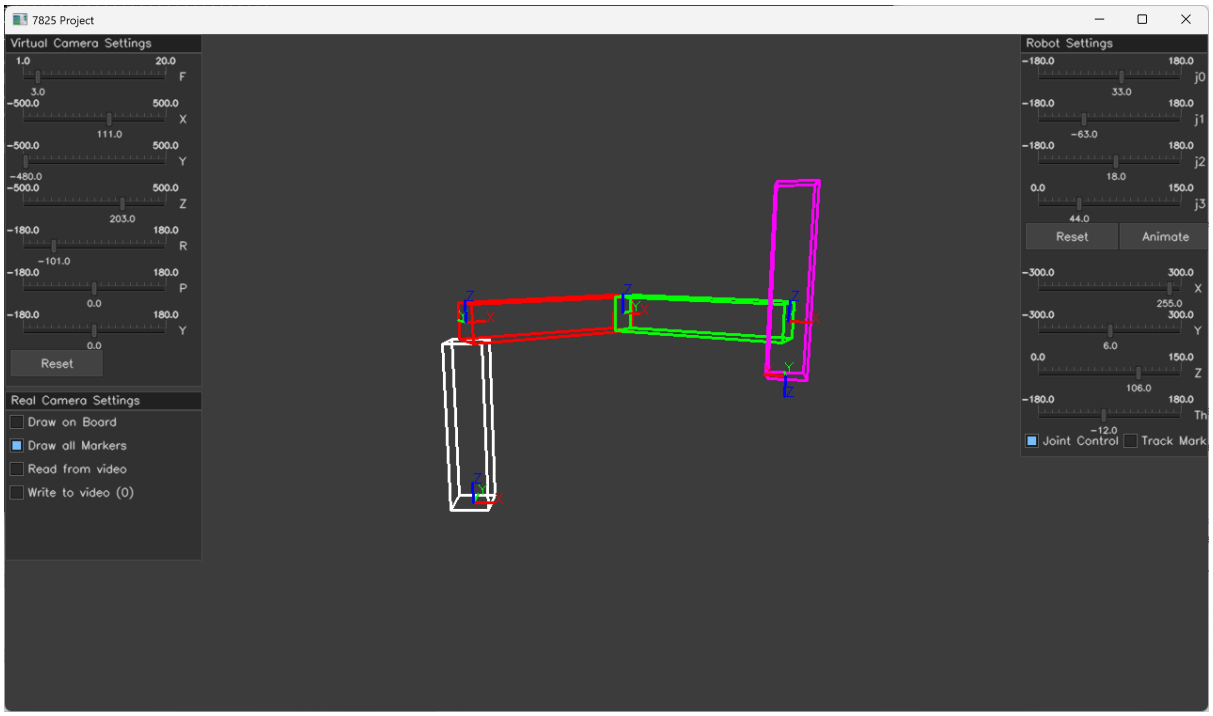
2.1 SCARA Robot Part A

The ELEX7825 C++ Template project provides a matrix representation of a box (8x 3D points).

A) Create a CRobot object that implements a SCARA robot with 4 joints, all links are 15 cm in length and 3 cm in height and depth. The origin of the box is the center of the face assigned to the joint. The coordinates are applied as described in the lecture notes. X axis (red), Y axis (green) and Z axis (blue). See figure below for illustration. A **draw** method should display the robot using the virtual camera.
After completing the robot, placed it on a 13.5 cm tall pedestal (15 cm minus ½ the link box height of 3 cm) so the prismatic joint at 0 length is located at 15 cm in the world coordinates.

B) Add a ***fkin*** method to the CRobot object which computes the end effector pose (output) based on the forward kinematics of the joint angles (input). This is basically the same operation as in Step A above except no need to calculate the intermediate joint poses.

C) Create a GUI interface with a slider to control each joint and display the current end effector pose (using the ***fkin*** method). Display a graphical representation of the manipulator for each set of joint inputs. Add a reset button. Note only one `cvui::update();` should be called for pushbuttons to work correctly.



2.2 SCARA Robot Part B

A) Add a button which animates the arm using a loop to increments joint 1 from 0 to +180 to -180 to 0, then joint 2 from 0 to +180 to -180 then to 0 degrees, then joint 3 from 0 to 360 degrees, then the prismatic joint from 0 cm to +15 cm.

B) Add the ability to draw the robot located at the origin of the ArUco board rather than the virtual camera. Joint manipulation should still work along with the animation button.

